AI-Based Study Planner Using Python

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AI-Based Study Planner Using Python

A rule-based intelligent system for optimizing daily study schedules through priority-driven time allocation

Project Overview

Course

Fundamentals in AI and ML (CSA2001)

Implementation

Python-based command line application

Approach

Rule-based AI prioritization algorithm

This project demonstrates practical application of AI concepts through an intelligent study scheduling system that optimizes time allocation based on subject difficulty, importance, and exam proximity.

Why This Matters

The Challenge

Students in today's academic environment struggle with inefficient time management across multiple subjects. Traditional planning methods lack adaptability and personalization.

The Solution

An intelligent system that uses data-driven logic to allocate study time effectively, considering subject difficulty, importance, and exam proximity.

How It Works

User Input

Subject details collected

Priority Calculation

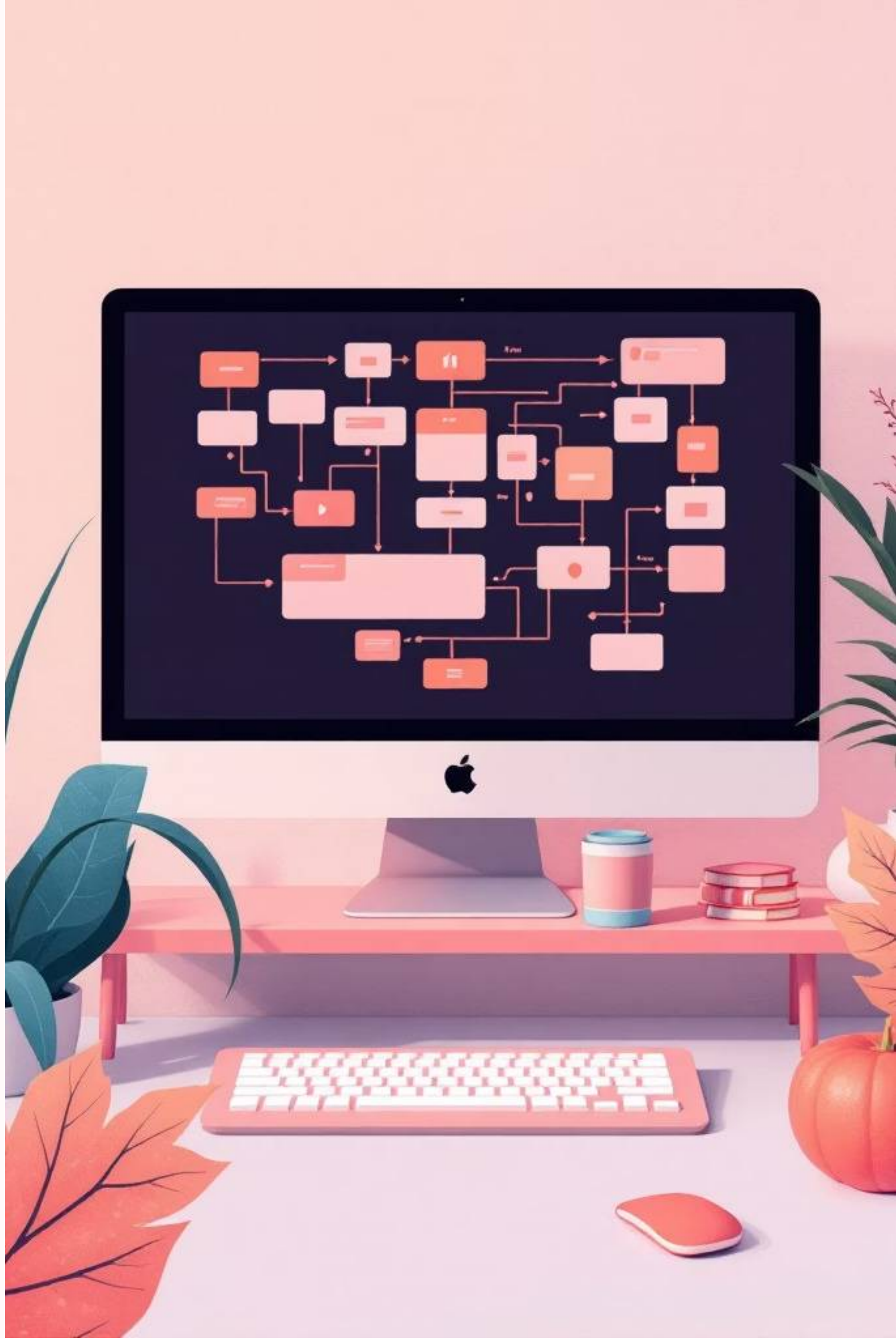
Score computed for each subject

Time Allocation

Hours distributed proportionally

Daily Schedule

Optimized plan displayed



Priority Calculation Formula

1

Gather Parameters

Difficulty level (1-10), importance rating (1-10), and days remaining until exam

2

Apply Weighting

Different factors weighted differently: difficulty and importance each contribute 40%, days remaining contributes 20%

3

Calculate Score

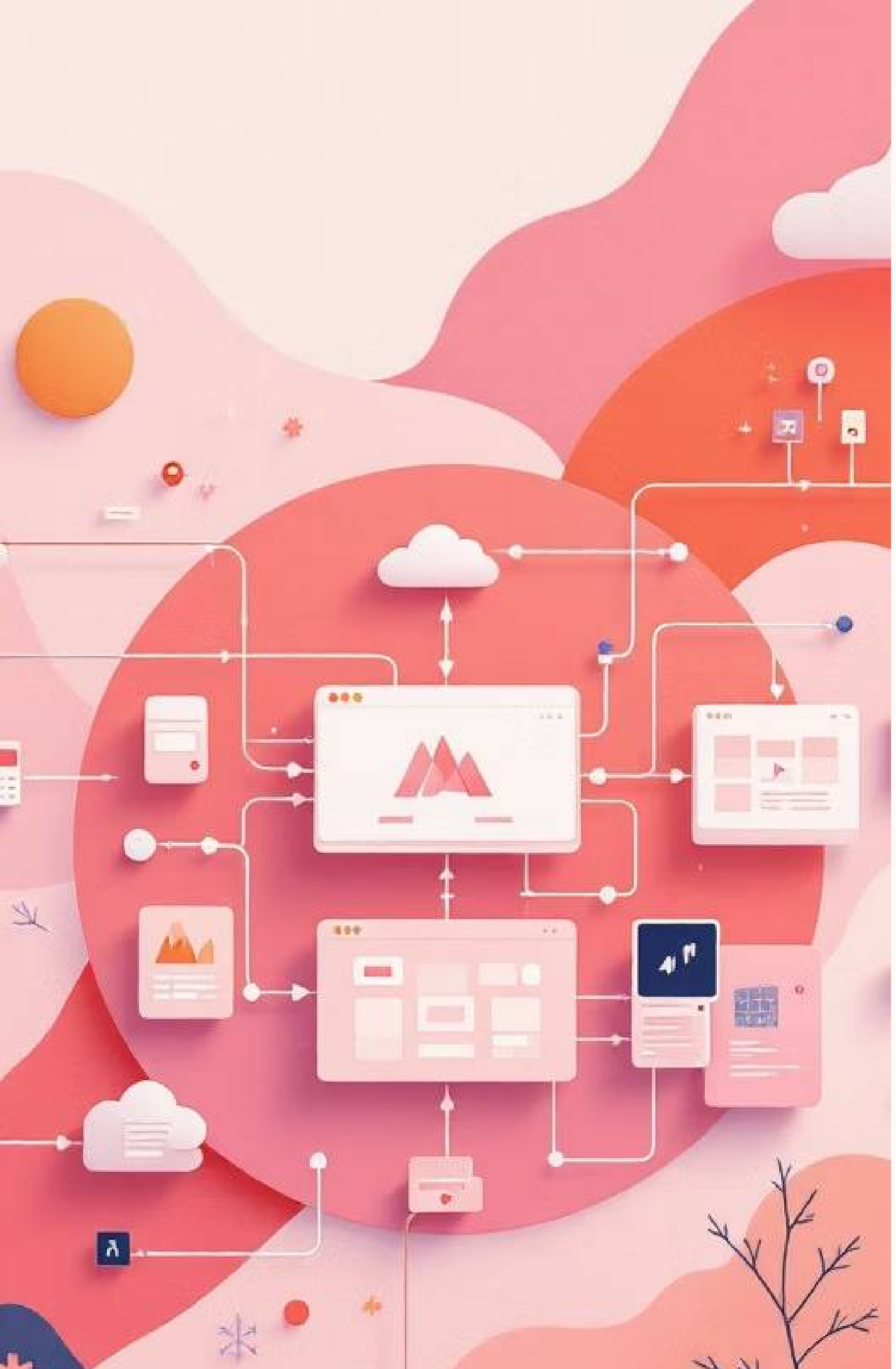
priority = (difficulty × 0.4) + (importance × 0.4) + (1/days_left × 0.2)

4

Sort & Allocate

Subjects sorted by priority, time distributed proportionally based on scores

Key Insight: The reciprocal of days remaining ensures urgent subjects receive higher priority as exams approach.



System Architecture

01

Input Module

Collects subject details from user

02

Processing Module

Calculates priority scores

03

Scheduling Module

Allocates time based on priorities

04

Output Module

Displays optimized study plan

Sample Output

Mathematics

2.8 hours

Difficulty: 8 | Importance: 9 | Days: 5

Physics

2.3 hours

Difficulty: 7 | Importance: 8 | Days: 7

English

0.9 hours

Difficulty: 5 | Importance: 6 | Days: 15

The system generates a daily study schedule showing hours allocated to each subject based on calculated priority scores.

Key Achievements

Improved Time Management

Automated allocation reduces planning time from hours to minutes

Balanced Coverage

Ensures no subject is neglected through proportional distribution

Reduced Effort

Eliminates decision fatigue with data-driven recommendations

AI Foundation

Practical demonstration of intelligent system concepts

Future Enhancements

Machine Learning Integration

Train models on student performance data to predict optimal study patterns

GUI Development

Build intuitive interfaces using Tkinter, PyQt, or web frameworks

Data Persistence

Add database integration to track progress and learning patterns

AI Assistant

Implement chatbot for interactive study guidance and feedback

Conclusion

This AI-based study planner successfully demonstrates how simple algorithms can significantly improve study efficiency and time management. By applying rule-based AI logic to a common student challenge, the project bridges theoretical AI concepts with practical application.

Key Contributions:

- Working prototype of intelligent scheduling system
- Clear demonstration of priority-based algorithms
- Foundation for advanced AI/ML integration
- Practical educational tool with real-world applicability

Next Steps: Expand to machine learning models, add user interface, and integrate with learning platforms for comprehensive study assistance.